

DRE Hackathon · Problem Statement Book

# Join the DRE revolution

Ten field-tested problem statements on decentralized renewable energy across Andhra Pradesh's coastal, agricultural and tribal belts — for teams building solar, microgrid and IoT solutions that hold up outside a lab.

Andhra Pradesh · North Coastal Districts

In partnership with



Ten problem statements. Four themes — agriculture, aquaculture, urban & industrial MSMEs, and tribal livelihoods. One shared constraint: build something a cooperative or a plant manager could put to use next season, not just next demo day.

**49%**

of global installed power capacity is now renewable

**55.3 GW**

non-fossil capacity India added last fiscal year

**9.1%**

of India's renewable potential sits in Andhra Pradesh

**<6%**

of the state's 167 GW potential is in use today

## The global commercial landscape is undergoing a silent structural shift — and the grid, not the panel, is where it's being felt first.

For decades, micro, small and medium enterprises (MSMEs) and large-scale industries alike have relied on centralized grids to fuel their daily operational needs. That top-down dependency introduces vulnerabilities of its own: grid instability, thermal overloads, fluctuating commercial tariffs and a heavy carbon footprint.

Today, Decentralized Renewable Energy (DRE) — rooftop solar, local microgrids, IoT-enabled productive-use appliances — is moving out of the backup-generator role. It is becoming a primary driver of corporate productivity and operational resilience, not a hedge against outages.

The shift is structural, and it is fast. Renewables now account for **49%** of all global installed power capacity, and almost all of the net additions — **96.8%** of them — came from variable solar and wind. Decentralization has stopped being a forecast and become the current baseline.

What that means in practice, for the enterprises this hackathon is built around, is the subject of the next two pages: a region with real potential, real constraints, and ten problems worth solving.

## Andhra Pradesh has the resource. The bottleneck is everything around it.

India installed 55.3 GW of non-fossil capacity last fiscal year and now ranks third globally in installed renewable capacity. Andhra Pradesh is among the country's top states, holding 9.1% of the nation's entire renewable energy generation potential. Despite a technical ceiling of roughly 167 GW, less than 6% of that capacity is in use today.

**167 GW**

theoretical renewable generation potential in Andhra Pradesh

**3rd**

India's global rank by installed renewable capacity

**96.8%**

of new global capacity came from solar and wind alone

The state's policy identifies wind and pumped hydro storage as the two main sources of renewable energy in the north coastal districts, alongside solar. But as the government pushes to meet its policy goals, three ground realities keep surfacing — and each one needs a different kind of solution.

- 1 Reverse power is stressing the grid it feeds.** Rapid, uncoordinated commercial solar adoption in dense corridors like Gajuwaka or Atchutapuram pushes heavy, volatile reverse power into localized distribution networks during sunny afternoons — causing voltage spikes that stress substation transformers.
- 2 Terrain keeps the grid — and repair crews — away.** Hilly topography and dense forest cover make expanding the central grid physically and financially prohibitive. Tribal processing cooperatives handling coffee, pepper and millets in Paderu or Chintapalli are well suited to isolated DRE machinery, but when an inverter or motor faults, the lack of local diagnostics or skilled technicians turns a fault into weeks of downtime and spoiled harvests.
- 3 Heavy loads are hard on light infrastructure.** Agricultural and shrimp-farming enterprises need continuous power for cold storage and pond aerators. But the inrush current from heavy inductive machinery trips low-cost rural solar microgrids and accelerates battery degradation — making solar-plus-storage a financially difficult call for local cooperatives.

These realities translate into ten concrete problem statements, grouped into four themes. Pick one, and build a solution that survives contact with the field.

Theme A

# Agriculture & horticulture value chains

Jaggery, coffee, cocoa and secondary produce — where heat, moisture and timing decide whether a harvest becomes revenue or waste.

## 1

## Solar-assisted jaggery processing optimization & heat tracking

Value chain: Sugarcane and jaggery

Local micro-enterprises struggle with inconsistent quality during sugarcane juice boiling, caused by uneven biomass or solar-hybrid heat distribution. Overheating or underheating ruins the jaggery grade.

**Build:** an IoT simulator and heat-optimization dashboard that monitors temperature sensors along the clarifying and boiling pans.

### Deliverables

A mobile dashboard showing heat thresholds, a predictive timer alerting the worker when to pour the juice, and an automated ledger tracking batch grade against temperature consistency.

## 2

## Solar dryer predictive yield & moisture tracker for cocoa and coffee

Value chain: Coffee cultivation and cocoa beans

Most farmers dry their produce under open sun, exposed to shifting local weather. Even with solar tunnel dryers or covered collective drying infrastructure, operators are still guessing when beans are perfectly dry — pulled too early they mold, left too long they lose weight and profit.

**Build:** a data-driven moisture evaporation predictor app that combines ambient weather APIs (humidity, cloud cover) with internal dryer sensors to calculate drying curves.

### Deliverables

A real-time interface showing estimated remaining dry time, automated WhatsApp or SMS alerts when target moisture is reached (e.g. 7% for coffee), and a historic drying log for quality certification.

### 3

## DRE-powered jackfruit & banana flower processing scheduler

Value chain: Secondary produce processing

Tribal women's cooperatives run solar-powered cutting, slicing and dehydration machinery that must operate immediately after harvest — but variable weather can drain the solar microgrid batteries mid-operation.

**Build:** a production scheduling and load-balancing web app for decentralized food processing hubs.

### Deliverables

An algorithm that reads daily harvest volumes and solar generation forecasts to recommend the best hours to run heavy slicing machines versus low-power dehydrators, avoiding battery depletion.

Theme B

# Aquaculture & cold chain resilience

Shrimp hatcheries, cold storage and shared cooling infrastructure — where a single power dip can spoil a season's work.

## 4

## Smart thermal load shifting for shrimp hatcheries & cold storage

Value chain: Shrimp and prawn farming

Moving rural cold rooms and hatchery aerators fully to solar-plus-storage mini-grids is expensive because of high battery costs — and hatcheries face immediate crop spoilage if power drops abruptly.

**Build:** an intelligent thermal and aeration balancing engine that dynamically adjusts energy use around peak sunlight.

### Deliverables

A prototype that safely "over-cools" storage units during peak sunlight hours — storing thermal energy to cut night-time draw — paired with an emergency battery-saving protocol for aerators on low-light days.

## 5

## Pay-as-you-go shared solar cooling hub wallet

Value chain: Marine fisheries and fresh vegetables

Smallholder farmers and fishermen can't afford their own solar refrigerators. Shared community cooling hubs exist, but manual slot booking and payment collection stay chaotic.

**Build:** a UPI-integrated PAYG booking and micro-billing mobile app for shared solar cooling grids.

### Deliverables

A consumer app where users scan a QR code on a specific cooling locker, unlock it via digital handshake, pay per hour or per kilogram through a mock UPI gateway, and view a live dashboard of diesel emissions the village has saved.

Theme C

# Urban infrastructure & industrial MSMEs

Rooftop clusters and manufacturing floors — where solar's success is starting to strain the grid that hosts it.

## 6

## Commercial rooftop solar cluster feasibility & grid transformer safety tool

Value chain: Industrial MSMEs, pharma clusters, large commercial establishments

Rapid, uncoordinated rooftop solar adoption by commercial buildings pushes heavy reverse power into local distribution transformers during sunny afternoons — risking grid failure.

**Build:** a geospatial solar penetration and transformer hosting-capacity simulator for local utility networks.

### Deliverables

An interactive map (Streamlit or Leaflet) that plots local transformers, lets a user drop a pin to simulate a new rooftop system, and instantly returns a grid safety impact score with a capacity threshold warning.

## 7

## Dynamic time-of-use tariff engine for manufacturing clusters

Value chain: Heavy fabrication, engineering and apparel manufacturing

Industrial units run heavy machinery during peak grid hours, driving up power costs, while rooftop solar arrays generate excess power that goes to waste whenever factories slow their operations.

**Build:** an AI-based power demand-shifting engine that matches production schedules to real-time solar availability.

### Deliverables

A scheduling tool where factory managers enter order deadlines and machinery power ratings, and the engine returns an optimized shift schedule that maximizes direct solar consumption and minimizes peak-hour grid draw.

Theme D

# Tribal livelihoods, eco-tourism & field operations

Remote valleys, homestays and field audits — where connectivity is the scarce resource, not power.

## 8

## Low-bandwidth remote diagnostic app for tribal machinery (offline-first)

Value chain: Non-timber forest produce — pepper thrashers, solar cleaners, millet hullers in remote Paderu / Chintapalli

Processing machinery runs in remote valleys with little or no internet connectivity. When an inverter or motor acts up, local operators have no way to diagnose it — and downtime stretches from hours into weeks.

**Build:** an offline-first IoT health diagnostic application that interfaces with machinery over local Bluetooth or Wi-Fi hotspots.

### Deliverables

An app that reads machine fault codes locally without internet, walks the operator through simple repairs via a lightweight embedded diagnostic tree, and packages logs into a compressed text format that syncs automatically over SMS once a cell tower is found.

## 9

## Eco-tourism microgrid sizing calculator & guest visibility dashboard

Value chain: Experiential tribal homestays and eco-resorts

Homestay owners want to go fully green on solar and micro-hydro, but struggle to size how many panels or batteries they actually need without overpaying for a commercial configuration.

**Build:** an eco-resort microgrid planner and green metric tracker.

### Deliverables

A simple, bilingual (Telugu / English) web form where a homestay owner enters room count and appliances to receive an optimal DRE blueprint, plus a guest-facing display showing live metrics — for example, "your stay today ran on 100% solar power, saving 12 kg of CO<sub>2</sub>."

## AI field assistant for DRE energy audits & livelihood assessment

Value chain: Rural enterprise mobilization and NGO field operations

Field coordinators auditing energy use across hundreds of tribal villages spend hours filling out forms by hand just to check whether a solar pump or food processor is viable for a given community cluster.

**Build:** an AI-powered, voice-based energy audit and baseline assistant that works fluently in regional languages.

### Deliverables

A voice-to-text mobile tool where field agents speak naturally in Telugu or English to record village data — active shops, current diesel costs, daily operating hours — which a lightweight NLP model turns directly into a standardized technical and economic feasibility summary.

# Problem-to-bottleneck mapping

Each problem statement traces back to one global bottleneck in the DRE transition, and one core technical focus a team is expected to demonstrate.

| PS   | Hackathon problem statement    | Primary global bottleneck addressed       | Core technical focus for participants                                                        |
|------|--------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|
| PS 1 | Solar-biomass controller       | Industrial heat decarbonization gap       | Automated electromechanical air/fluid dampers balancing variable heat sources.               |
| PS 2 | Solar dryer moisture tracker   | Agricultural metrology & quality risk     | Data-driven drying curves syncing ambient weather APIs with moisture thresholds.             |
| PS 3 | Processing scheduler           | Production & intermittency mismatch       | Production load-balancing algorithms matching machinery run times to solar generation maps.  |
| PS 4 | VFD aerator & thermal storage  | Chemical battery capacity & lifetime cost | Soft-starting high inrush induction motors and leveraging thermodynamic thermal-mass shifts. |
| PS 5 | PAYG shared cooling hub wallet | Upfront capital expense (CapEx) barrier   | UPI-integrated micro-billing and dynamic slot reservation architecture (CapEx to OpEx).      |
| PS 6 | Rooftop solar grid safety tool | One-way distribution grid volatility      | Geospatial transformer hosting-capacity simulation and reverse power throttling.             |
| PS 7 | Dynamic ToU tariff engine      | Demand-side flexibility deficit           | Production scheduling tools using time-of-use pricing models to absorb solar abundance.      |

| PS           | Hackathon problem statement   | Primary global bottleneck addressed      | Core technical focus for participants                                                               |
|--------------|-------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>PS 8</b>  | Low-bandwidth diagnostic app  | "Last-mile" diagnostic & maintenance gap | Offline-first local Bluetooth diagnostic trees and compressed text-over-SMS logging.                |
| <b>PS 9</b>  | Eco-tourism sizing calculator | Technical sizing & planning illiteracy   | Sizing optimization forms converting basic appliance inventories into clean-energy blueprints.      |
| <b>PS 10</b> | AI field assistant            | Feasibility study & energy audit delays  | Voice-to-text NLP processing for instant, standardized rural economic profiles in regional tongues. |

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Pick one problem statement, one theme, one region-specific constraint — and build a solution that a jaggery cooperative in Chintapalli, or a plant manager in Atchutapuram, could actually put to use.